

Aminu Musa

Languages: fluency in English & Hausa | **Nationality:** Nigerian | **Address:** B10-1B Malam Inuwa Dutse Housing Estate, Dutse, Nigeria | **Phone:** 07033120654 |

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EDUCATION

Bayero University Kano

Kano, Nigeria

Master of Science in Computer Science

October 2020 – February 2025

Ahmadu Bello University

Zaria, Nigeria

Postgraduate diploma in Education (PDE)

March 2020 – February 2021

Federal University Dutse

Jigawa, Nigeria

Bachelor of Science in Computer Science

January 2013 – November 2017

EMPLOYMENT

Assistant Lecturer

June 2020 – Present

Federal University Dutse

Dutse, Nigeria

- Developed lecture materials for programming course such as Java, python, and C++
- Teach undergraduate courses such as: Theory of Computation, Introduction to programming, and Survey of programming languages
- Conduct examinations and. Mark student assessments Chaired the committee that designed and facilitated the department welfare
- Mentor and supervised more than 50 undergraduate students' final year project
- Organized and led machine learning classes/tutorials for over 500 Students for 3 years
- Conduct collaborative research with academia and industry on Artificial Intelligence, computer vision in healthcare

Graduate Assistant

July 2021 – February 2022

Jigawa State College of Education Gumel (JSCOE)

Jigawa, Nigeria

- Taught CS 0124: Introduction to programming, CS124: Introduction to computer science, college-level courses to over 400 students with great feedback from the students as well as the faculty
- Conduct grading for continuous assessments, examinations, and other assignments

System Analyst

July 2018 – June 2019

Al-Qalam University

Katsina, Nigeria

- Manage postgraduate admission portal, proctored tests, and provided sectarian duties such typing, printing, troubleshooting
- Manage network infrastructures at school of postgraduate

Software Engineer

November 2017 – April 2019

Digital Development Hub (DD-HUB)

Abuja, Nigeria

- Handle backed programming for a team that developed a mobile app for Universal Basic Education (UBEC) project monitoring
- Developed an app (Autofix) using NodeJS, Mango DB database and VueJS for national automotive agency
- Developed several web applications using php, JavaScript, and MySQL databases for several clients

Data Analyst

February 2015 – August 2015

National Population Commission (NPC)

Kano, Nigeria

- Collected, interpreted over 3 different survey datasets, and reported the results back to the project manager
- Analyzed survey data using SPSS for out of school children NEEDs assessment
- Managed and analyzed and interpreted different survey datasets using R
- Wrote weekly progress reports on milestones achieved

CERTIFICATION AND AWARDS

- TensorFlow Developer Certificate August 2023
- NITDA AI Professional Certificate April 2024
- AWS Machine learning Foundations Certificate November 2022
- Nvidia DLI Fundamental of Deep learning Certificate January 2022
- University of Washington, Computer Networks Certificate January 2013
- Google Research Explore CSR grant awardee for the Computing Research Acceleration Initiative (CRAI) program 2022, 2023, 2024
- Kaggle ML Research spotlight Prize winner 2022.
- TensorFlow University RFP Awards, awardee for introduction to Machine learning classes (ML-101) program at Federal University Dutse 2021, 2022, 2023
- Medical Image Computing and Computer Assisted Interventions Society (MICCAI) Student Participation Award 2022, 2024
- Petroleum Technology Development Fund (PTDF) In-Country Postgraduate Scholarship Award 2020 winner
- Federal Government Scholarship Award (FSB) Undergraduate Merit 2016 awardee

PROJECTS

An Intelligent Plant Diseases Detection System 2022

- Developed a deep learning-based system to detect 38+ diseases across 8 plant species, improving early diagnosis for farmers.
- Implemented using Python, TensorFlow, Keras, and Scikit-learn; deployed with Flask as web app for real-time access
- Enabled timely intervention to reduce crop loss and support sustainable agriculture.

Large Vision Language Model (VLM) For Radiology Report 2025

- Designed and fine-tuned a vision-language model (VLM) to automatically generate diagnostic radiology reports from chest X-rays.
- Integrated supervised adversarial domain adaptation to address cross-population domain shift and improve generalization on Nigerian clinical data.
- Utilized Python, PyTorch, and Hugging Face Transformers; evaluated performance with BLEU/METEOR and CheXpert clinical metrics.
- Advanced equitable AI in healthcare by enabling accurate reporting for underrepresented populations.

A web application for civil registration at Natl. Pop. Comm 2017

- Built a full-stack web application for managing civil registrations, improving record accuracy and accessibility.
- Developed with PHP, MySQL, JavaScript, HTML, and CSS, ensuring secure data handling and user-friendly interfaces.
- Streamlined registration workflows and reduced manual errors, supporting efficient service delivery at the National Population Commission.

MEMBERSHIPS

- Member Black in AI
- Member Nigerian Computer Society (NCS),
- Medical Image Computing and Computer Assisted Interventions Society (MICCAI) Society Member
- Member MICCAI Student Board (MSB)

TECHNICAL SKILLS

Languages: Python, Java and C

Frameworks: Scikit-Learn, TensorFlow, Pytorch

Developer Tools: Git/GitHub, Amazon Web Services (AWS), Google Cloud Platform (GCP)

PUBLICATIONS

- Musa, A., Ibrahim Adamu, M., Kakudi, H.A., Hernandez, M., Lawal, Y.
Analyzing Cross-Population Domain Shift in Chest X-Ray Image Classification and Mitigating the Gap with Deep Supervised Domain Adaptation.
In: Medical Image Computing and Computer Assisted Intervention – MICCAI . Morocco 2024, LNCS vol. 15003, pp. 585–59
- Musa, A., Prasad, R., Hernández, M.
Addressing Cross-Population Domain Shift in Chest X-Ray Classification through Supervised Adversarial Domain Adaptation.
Scientific Reports 15, Article 11383 (2025).
- Musa, A., Aliyu, F.
Performance Evaluation of MLP and RBF for handwritten digit recognition.
Presented at IEEE NigeriaComputConf. Nigeria, 2019.
- Musa, A., Hamada, M., Aliyu, F., Hassan, M.
An Intelligent Plant Disease Detection System for Smart Hydroponics Using CNN.
Presented at IEEE 14th International Symposium on Embedded Multicore/Many-core Systems-on-Chip (MCSoc), Singapore, 2021.

Full list at [Google Scholar](#).

NON-TECHNICAL SKILLS

Languages: English, Hausa

HOBBIES

Travelling, Sport activities

REFEREES

- Dr. Farouq Muhammad Aliyu
King Fahd University of Petroleum and Minerals (KFUPM).
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- Prof. Abubakar Muhammad Miyim
Federal University Dutse (FUD), Jigawa State.
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